

Syllabus, Timeline & Lab Requirement

How to read this annexure

- **Three programmes, two module libraries.** A **1-year PG Diploma** and a **6-month Training Programme** are cut from one graduate-tier library. A **6-month post-ITI Technician Programme** runs on its own library. Hours in Part 1a are shown as [Diploma / Training](#).
- **Cohort size for all quantities: 30 students per programme.** Working groups: **15 pairs** for bench work, **6 teams of 5** for live-cell work.
- **Greenfield.** This assumes **BGR Neo holds no robots or lab assets today**. Everything required is listed. Anything already owned is netted off at procurement.
- **No prices.** Each line carries a relative cost signal — **LOW · MEDIUM · HIGH**. Costing is finance's to complete.
- **Revision 2** applies the corrections in [curriculum/VALIDATION-REPORT.md](#). The diploma is now **1,200 h / 40 credits** (UGC's cap for a PG Diploma), a **PLC module** has been added to both tiers, the specialization block is rebalanced so the station rotation is schedulable, and an **assessment cell** is ring-fenced.

PART 1A — SYLLABUS · GRADUATE TIER

Awarding body: Loyola College, Chennai (autonomous, arts & science, University of Madras, UGC-regulated) for the PG Diploma. The 6-month Training Programme carries a **BGR Neo CoE certificate plus vendor certificates** — *not* an NSQF/QP certificate, because both candidate Qualification Packs set a minimum job-entry age of 24 and exclude a fresh-graduate cohort.

1.1 Structure at a glance

Block	Title	Diploma	Training
F	Foundations	230 h	90 h
R	Core Robotics	310 h	200 h
D	AI Depth	240 h	140 h
S	Industrial Specialization (<i>identical in both</i>)	260 h	260 h
C	Capstone & Employability	160 h	120 h
	TOTAL	1,200 h	810 h
	<i>Credits @ 30 notional h</i>	40 — a UGC PG Diploma exactly	27

1,200 h is **notional learning hours** per NCrF — *directed contact plus supervised self-study* — *not contact hours alone*.

1.2 Block F — Foundations · 230 h / 90 h

#	Module	Sub-modules	Diploma	Training
F1	Engineering Maths for AI	Linear algebra & transforms · Probability & statistics · Calculus & numerical methods · Optimization basics	80	30
F2	Programming	Python for engineers · Data structures & algorithms · C++ essentials for robotics · Linux, Git & dev workflow	90	30
F3	Classical Machine Learning	Supervised learning · Unsupervised learning & model evaluation	30	0
F4	Industrial Context, Safety & Standards	Manufacturing processes & factory layout · ISO 10218-1/-2, ISO/TS 15066, IEC 60204-1, ISO 13855, ISO 13849-1 & risk assessment · Shop-floor discipline (LOTO, PPE, 5S, permit-to-work)	30	30

1.3 Block R — Core Robotics · 310 h / 200 h

#	Module	Sub-modules	Diploma	Training
R1	Robot Kinematics & Dynamics	Coordinate frames & homogeneous transforms · Forward & inverse kinematics · Jacobians, singularities & workspace · Dynamics, payload & reach sizing	60	40
R2	Industrial Robot Programming	Teach-pendant programming (jogging, frames, TCP, I/O) · Motion types & production program structure · Offline programming & simulation · Multi-vendor: FANUC / ABB / KUKA / UR	60	60
R3	ROS 2 + MoveIt 2	ROS 2 core (nodes, topics, services, actions) · URDF, TF2, RViz & launch · MoveIt 2 planning & pick-place · ros2_control	60	30
R4	Feedback Control Fundamentals	Modelling & transfer functions · PID design & tuning · Servo drives, encoders & motion profiles · State-space intro	40	20
R5	Motion Planning & Mobile Robotics	Path & trajectory planning · SLAM & localization basics · Nav2 & AMR fleet basics (VDA 5050)	30	10
R6	Industrial Controls & PLC (<i>new in Rev 2</i>)	Ladder logic & function blocks · PLC ↔ robot handshaking · Industrial networks (Ethernet/IP, PROFINET, Modbus, OPC-UA) · HMI design · Safety circuits & validation	60	40

R6 exists because industrial cells run PLCs and vendor stacks, not ROS 2. Revision 1 buried all controls content inside S3, where 60 h had to cover PLC programming, four fieldbuses, HMI design and safety validation — while the lab bought PLC trainers that no module had hours to use. R3 was reduced to pay for it.

1.4 Block D — AI Depth · 240 h / 140 h

#	Module	Sub-modules	Diploma	Training
D1	Deep Learning	Neural network fundamentals & training · CNNs & image models · Transfer learning & small-data regimes · Deployment & edge inference	80	50
D2	Machine Vision for Manufacturing	Industrial imaging: optics, lighting & camera selection · Classical vision & metrology · Deep-learning defect detection · 3D vision, point clouds & 6-DoF pose for bin picking · Hand-eye calibration	90	70
—	Elective pool — diploma students choose 2 of 4		70	20

Elective	Content	Hours
D3	Reinforcement & imitation learning · sim-to-real & domain randomization	35
D4	Industrial data & time series · OPC-UA/MQTT & historians · anomaly detection · dashboards	35

Elective	Content	Hours
D5	Agentic AI & LLM applications for manufacturing — MES/ERP agents, conversational HMI, guardrails	35
D6	Simulation & digital-twin depth — physics engines, synthetic data, real2sim	35

The Training Programme takes **D4 only, at 20 h**. D3 is tagged droppable and is genuinely dropped from it.

1.5 Block S — Industrial Specialization · 260 h in both programmes

Identical in the Diploma and the Training Programme. Delivered on live industrial cells. Rebalanced in Revision 2 to near-equal dwell so six teams can actually round-robin six stations — the previous 50/50/60/40/30/30 split could not.

#	Module	Sub-modules	Hours
S1	Vision-Guided Pick & Place on a Live Cell	Cell design & cycle-time budgeting · Vision-guided part localization · Gripper/EOAT selection & integration · Build, run & optimize	45
S2	Welding / Dispensing OLP + Digital Twin	Process fundamentals · Offline programming & path generation · Digital-twin build & simulation validation · Download, touch-up & production proving	45
S3	Cell Integration	Robot ↔ PLC handshaking · Industrial networks in practice · HMI/SCADA · Safety circuit design, guarding & validation	45
S4	Predictive Maintenance & Quality Analytics	Condition monitoring (vibration, current, thermal) · Failure modes & remaining-useful-life · SPC & quality analytics on live line data	40
S5	Operator-Facing AI (MES/ERP + Conversational HMI)	MES/ERP data flows & production KPIs · AI assistants for SOPs and diagnostics · Guardrails, validation & deployment	40
S6	Calibration, Commissioning & Handover	Robot mastering, TCP & payload calibration (ISO 9283) · FAT/SAT protocols · Documentation, as-built drawings & operator training	45

S6's operator-training component has a real audience — the ITI cohort. See Part 2.4.

1.6 Block C — Capstone & Employability · 160 h / 120 h

#	Module	Sub-modules	Diploma	Training
C1	Capstone Project	Problem definition & scoping · Design, simulation & digital twin · Build & integration on the live cell · Testing, validation & optimization · Defence	130	90
C2	Portfolio & Employability — mapped to DGT/VSQ/N0102	Technical documentation & drawings · Customer communication & site conduct · Portfolio build & interview preparation	30	30

PART 1B — SYLLABUS · TECHNICIAN TIER (POST-ITI)

Two separately certified NSQF Level 4 awards of 450 h each. Full design, entry bands, certification route and rationale in [curriculum/iti-operator-technician-programme.md](#).

1.7 Structure at a glance

Award	Trade hours	+ Employability	Total	Weeks
Pre-gate bridge (Stream A / Stream B)	—	—	60 / 120	2–4
Tier 1 — Automation Cell Operator	420	30	450	13
Tier 2 — Automation Uptime Technician	420	30	450	13
	840	60	900	26

450 h sits inside the NSQF Level 4 notional band (390–480 h), equals 15 credits exactly, and falls inside PMKVY's 300–600 h Short-Term Training band — so each tier is independently fundable. A single 760 h award, as originally drafted, fitted no band at all.

1.8 Tier 1 — OPERATE · 420 h + 30 h Employability

#	Module	Sub-modules	Hours
01	Safety & the Cage	Shop-floor safety, PPE, LOTO, permit-to-work · Safety circuit: light curtains, scanners, interlocks, E-stop categories, muting · Safe entry, recovery, restart · ISO 10218-1/-2, ISO/TS 15066, IEC 60204-1, ISO 13855, IEC 61496, ISO 13849-1 / IEC 62061	60
02	Anatomy of an Automated Cell	The seven fault domains · Reading a cell: layout, I/O list, sequence of operations · Cycle logic	50
03	Cell Operation & HMI	Start-up, shutdown, mode selection · HMI navigation · Changeover & first-piece verification · Cycle and quality monitoring	60
04	Robot Handling for Technicians	Robot safety, jogging, frames & TCP · Home/park positions; mid-cycle stop recovery · Program selection, backup, restore, touch-up within authority	60
05	Material Handling & Flow	Conveyors, transfers, jams · Palletizing & depalletizing: layer patterns, slip sheets, wrap, pallet quality · Machine tending · Dock discipline & AGV/AMR interaction	50
06	Reading the Machine: Sensors, Signals & Alarms	Sensor types and what "no signal" means · Barcode, RFID, vision pass/fail · Alarm codes	60
07	The Fault Loop — Drills	The six-step loop · Seeded-fault drills across all seven domains · Timed recovery assessment	40
08	Measurement & First-Piece Quality	Gauging, go/no-go, basic metrology · Reading an SPC chart · First-piece and in-process checks	20
09	Environment, Energy & Waste	E-waste & electronics disposal · Coolant and oil handling · The cost of a compressed-air leak	20
ES	Employability Skills — DGT/VSQ/N0102	Communication, workplace conduct, digital & financial literacy	30

1.9 Tier 2 — MAINTAIN · 420 h + 30 h Employability

#	Module	Sub-modules	Hours
M1	Preventive & Autonomous Maintenance — TPM Jishu Hozen	PM schedules, lubrication, tension, torque · Gripper & vacuum system care · Vision optics care · CMMS: work orders, spares, history	60
M2	Electrical, Pneumatic & Hydraulic First-Line Repair	Panel layout & safe working; fuses, breakers, contactors, PSUs · Sensor replacement & alignment · Cable and continuity faults · Pneumatics: FRL, valves, cylinders, leaks · Hydraulics: pumps, valves, accumulators, filtration, contamination	90
M3	Mechanical First-Line Repair	Gripper/EOAT change & tool changers · Fixture and guide alignment · Belts, rollers, chains, couplings · Bearing & vibration first checks	50
M4	Vision & Quality Subsystem Care	Why detection rate drops · Re-running calibration within authority · Model problem vs physical problem	40
M5	Structured Diagnosis & Escalation	Fault trees per domain; binary search · 5-Why & root cause · Writing an escalation report an engineer can act on · Spare-parts discipline	60
M6	Condition Monitoring & Working with the AI	Reading trends: vibration, current, temperature, cycle-time drift · What a predictive alert means and its limits	30
M7	Handover, Downtime & the Digital Shop Floor	Shift handover & logbook discipline · Downtime coding, OEE basics · MES / operator-AI assistant	30
M8	PLC & Control Logic Literacy — read-only (<i>new in Rev 2</i>)	Ladder & function-block basics · Programming software in monitor mode · Cross-referencing an alarm to logic · Forced I/O and status bits · Fieldbus diagnostic pages · Robot ↔ PLC handshake signals. Monitor mode only	60
ES	Employability Skills — DGT/VSQ/N0102	(<i>continued</i>)	30

M8 exists because two of the tier's own seven fault domains — *BRAIN* and *NERVES* — were previously untaught. The portability promise was 5/7 delivered.

PART 2 — TIMELINE

2.1 PG Diploma — 1 year · 48 teaching weeks + 4 examination/break weeks

Weeks	Phase	Blocks	Hours	~h/wk	Facility
1–10	Foundations	F1 · F2 · F3 · F4	230	23	Classroom + computer lab
11–20	Core robotics	R1 · R2 · R3 · R4 · R5 · R6	310	31	Computer lab + robotics bench + controls bench
21–30	AI depth	D1 · D2 · 2 electives	240	24	Computer lab (GPU)
31–40	Specialization	S1 · S2 · S3 · S4 · S5 · S6	260	26	LIVE CELLS
41–48	Capstone	C1 · C2	160	20	LIVE CELLS + classroom
Total			1,200		

Live-cell window: weeks 31–48 (18 weeks).

2.2 Engineer Training Programme — 6 months · 26 weeks

Weeks	Phase	Blocks	Hours	~h/wk	Facility
1–3	Bridge	F1 · F2 (refresher) · F4	90	30	Classroom
4–9	Core robotics	R1 · R2 · R3 · R4 · R5 · R6	200	33	Computer lab + benches
10–14	AI depth	D1 · D2 · D4	140	28	Computer lab (GPU)
15–22	Specialization	S1 · S2 · S3 · S4 · S5 · S6	260	33	LIVE CELLS
23–26	Capstone	C1 · C2	120	30	LIVE CELLS + classroom
Total			810		

Live-cell window: weeks 15–26 (12 weeks).

2.3 Technician Programme (post-ITI) — 6 months · 26 weeks

Weeks	Phase	Modules	Hours	~h/wk	Facility
1–2	Safety induction & the cage	O1	60	30	Classroom + safety rig
3–7	Operate: cell fundamentals	O2 · O3 · O4	170	34	Trainer cells
8–10	Operate: flow & signals	O5 · O6	110	37	Palletizing rig + trainer cells
11–12	Fault drills, measurement, environment	O7 · O8 · O9	80	40	LIVE CELLS + fault-injection rig
13	Employability + Tier 1 assessment	ES	30	30	LIVE CELLS
14–17	Maintain: preventive & first-line repair	M1 · M2	150	37	Benches + trainer cells
18–20	Maintain: control logic & mechanical	M8 · M3	110	37	LIVE CELLS
21–23	Maintain: vision & structured diagnosis	M4 · M5	100	33	LIVE CELLS
24–26	Condition monitoring, handover + final assessment	M6 · M7 · ES	90	30	LIVE CELLS
Total			900		

Live-cell window: weeks 11–13 and 18–26 (12 weeks).

2.4 The combined calendar — three cohorts, one cell hall

Start all three programmes in the **same week** and live-cell demand sequences rather than collides.

Weeks	On the live cells	Notes
1–10	(none)	All three cohorts on classroom, sim and trainer cells
11–13	ITI only	Fault drills + Tier 1 assessment
14	(none)	
15–17	Engineer training only	Specialization begins
18–26	Both — the Handover Model window	Engineers build and commission; technicians operate and maintain
27–30	(none)	Both 6-month cohorts have graduated
31–48	Diploma only	Specialization + capstone

33 of 52 weeks carry live-cell demand; only 9 are contended. Book cell access in **2-week blocks per team**, not continuously — real integration work is bursty, and block booking raises effective capacity roughly 3–4×.

The Handover Model (weeks 18–26) is milestone-gated — handovers occur at FAT, SAT and production-ramp boundaries, roughly 6–8 per cohort, plus daily shift-log handovers inside the ITI cohort's own shifts. It runs on a **ring-fenced training cell**, never on a customer-acceptance cell, and every milestone carries a staffed ISO 10218 / ISO-TS 15066 safety re-validation step.

2.5 Annual cohort capacity

Programme	Cohorts / yr	Students / yr	Live-cell weeks
PG Diploma	1	30	18
Engineer Training	1	30	12
Technician (post-ITI)	1	30	12
Total	3	90	33 of 52 weeks (9 contended)

A second cohort of either 6-month programme requires an additional cell or shift-based access.

2.6 The station rotation — how 30 students share the hardware

Block S is delivered as a **six-station rotation**: six teams of five rotate through six stations, so the Centre needs **one of each station, not six of everything**. Revision 2's rebalancing (45/45/45/40/40/45 h) makes the dwell near-equal, which is what makes a round-robin actually schedulable.

Station	Serves	Facility
1	S1 — Vision-guided pick & place	Live Cell 1
2	S2 — Welding / dispensing + OLP	Live Cell 2
3	S3 — Cell integration	Integration bench
4	S4 — Predictive maintenance	Condition-monitoring rig
5	S5 — Operator-facing AI	MES/AI workstation cluster
6	S6 — Calibration & commissioning	Calibration bench

PART 3 — LAB & EQUIPMENT REQUIREMENT

Sized for 30 students per programme, three programmes. Greenfield.

The capex logic depends on siting — and it may invert. The published TNSDC CoE EOI §12(1) states that TNSDC provides the physical space "in any of the government institutions or building," along with furniture, workstation PCs and connectivity, while the partner supplies the specialist equipment. **If the CoE is sited that way, "BGR Neo is mostly buying computers, not robots" reverses exactly** — Groups A and J shrink and Group C becomes the whole spend. Both scenarios must be costed before the money slide is written.

3.1 Group A — Compute, network & edge · MEDIUM

Serves the graduate tier. The technician tier needs almost none of it.

Item	Purpose	Ratio	Qty
GPU student workstation (RTX-class, ≥64 GB RAM)	Isaac Sim, deep learning, ROS 2, OLP	1 : 1	30
Spare student workstations	Failure cover	—	2
Instructor workstation + dual display	Demonstration & authoring	—	2
Shared training server (multi-GPU node)	Model training, batch jobs	1 : 30	1
NAS / storage server (~100 TB usable)	Datasets, projects, backups	1 : 30	1
NVIDIA Jetson Orin Nano / NX edge kit	On-robot inference labs	1 : 2	15
NVIDIA Jetson AGX Orin	Cell-side vision deployment	shared	3
MCU dev kits (STM32 / ESP32)	Embedded & micro-ROS labs	1 : 2	15
Rugged laptops	Shop-floor commissioning	1 : 5	6
Basic PCs (technician tier)	CMMS, HMI, documentation	1 : 3	10
Network: managed switches, rack, firewall, 10 GbE, UPS	Infrastructure	—	1 set
Classroom AV (display, camera, audio)	3 rooms	—	3 sets

3.2 Group B — Learning-tier robotics (bench) · LOW-MEDIUM

Item	Purpose	Ratio	Qty
Desktop 6-axis educational arm + gripper	R1, R3, O2-O4	1 : 5	6
Collaborative robot (UR5e/UR10e-class) on mobile bench, gripper	R2, R3, S1, O3-O4	1 : 10	3
Educational AMR (TurtleBot 4-class)	R5	1 : 5	6
Simulation seats (Isaac Sim, Gazebo, MuJoCo)	All sim work	1 : 1	free / open-source

3.3 Group C — Live industrial cells · HIGH

Cell 1 — Vision-guided pick & place / machine tending

Serves S1, C1 (graduate tier) and O5, M3 (technician tier) — the technician tier's machine-tending requirement is satisfied here rather than by a separate cell.

Item	Qty
Industrial 6-axis robot, ~10–20 kg payload (FANUC / ABB / KUKA / Yaskawa class), controller & pendant	1
Industrial 3D vision system (structured-light / bin-picking class)	1
2D machine-vision cameras + lenses	2
Machine-vision lighting (bar, ring, dome, backlight)	1 set
Servo gripper + vacuum EOAT + quick-change tool changer	1 set
Conveyor, part presentation fixtures, bins; machine-tool interface	1 set
Safety PLC + standard PLC + HMI panel	1 set
Perimeter fencing, interlocked gate, safety scanner, E-stops	1 set

Cell 2 — Welding / dispensing

Item	Qty
Industrial arc-welding robot (6-axis, welding package)	1
Welding power source + torch + wire feeder	1
2-axis positioner / turntable	1
Laser seam-tracking sensor	1
Fume extraction, weld screens, fire-rated enclosure	1 set
Safety circuit: fencing, interlocks, E-stops	1 set

Option: if welding capex or facility services constrain the launch, substitute a **sealant/adhesive dispensing cell** — same offline-programming and digital-twin outcomes, materially lower cost signal, no fume or fire services.

Cells 3 & 4 — Flexible capstone cells

Item	Qty
Collaborative robot (UR10e / FANUC CRX class), flexible table, quick-change tooling	2
Safety-rated laser scanner per cell	2
Reconfigurable fixturing / bench	2 sets

Cell 5 — Assessment cell (ring-fenced) · *new in Rev 2*

Item	Qty
Cobot or light industrial robot with vision, PLC, safety circuit and reconfigurable tooling	1

No cohort trains on this cell. The technician Tier 2 examination is an *unseen* cell with three seeded faults, and the graduate capstone requires an uncommitted cell — neither works if every cell is consumed by teaching. **Fallback if capex is tight:** run the assessment on a ring-fenced *configuration* — a different part, tooling, layout and fault set on Cell 3 or 4, locked during assessment windows (weeks 13, 24–26, 41–48). Weaker, but defensible; permanently sharing a trained-on cell is not.

Phase 2 (optional, year 2)

Item	Qty
Mobile manipulator — AMR with mounted cobot arm (C2 platform track)	1
Industrial AMR for fleet / VDA 5050 labs	1

3.4 Group D — Vision & sensing (teaching stock) · LOW-MEDIUM

Item	Ratio	Qty
Depth camera (Intel RealSense D435/D455-class)	1 : 2	15
Industrial machine-vision camera + lens (Basler / FLIR-class)	bench	8
Machine-vision lighting kits (bench)	1 : 8	4
2D LiDAR (RPLIDAR-class)	with AMRs	6
3D LiDAR (Livox Mid-360-class)	shared	2
IMU / encoder / sensor kits	1 : 5	6
Thermal imaging camera	shared	2

3.5 Group E — End effectors & tooling · MEDIUM

Item	Qty
2-finger servo gripper (Robotiq-class)	4
Vacuum / suction EOAT + venturi generator	3
6-axis force-torque sensor (ATI / Bota-class)	2
Quick-change tool changers	3
Custom fixture & jig fabrication allowance	1 lot

3.6 Group F — Industrial controls & integration · MEDIUM

Sized for two concurrent cohorts — the graduate tier's R6 and the technician tier's M8 both land in the controls bench, and the single-cohort sizing in Revision 1 became binding once M8 was added.

Item	Ratio	Qty
PLC trainer kit with digital/analog I/O (Siemens S7-1200 / AB CompactLogix class)	1 : 5 × 2 cohorts	12
PLC monitor-mode software seats (<i>new in Rev 2 — for M8</i>)	1 : 2	15
Safety PLC / safety relay trainer	1 : 10 × 2	6
HMI panel	1 : 5 × 2	12
VFD + motor + encoder trainer	1 : 5	6
Industrial network kit (Ethernet/IP, PROFINET, Modbus, OPC-UA gateway, managed switch)	1 : 5 × 2	12
Pneumatics trainer (valves, cylinders, FRL)	1 : 10	3
Hydraulic trainer boards (<i>new in Rev 2 — for M2</i>)	1 : 5	6

Item	Ratio	Qty
Electrical panel build kit (DIN rail, contactors, terminals, wiring)	1 : 5	6
Electrical fault-finding trainer boards	1 : 5	6
Pneumatic fault trainer boards	1 : 5	6

3.7 Group G — Safety infrastructure · MEDIUM (*non-negotiable*)

Item	Qty
Safety-rated laser scanner (SICK microScan3-class, Type 3 / PL d)	5
Light curtain pair	2
Safety relay / E-stop station / interlocked gate switch sets	5
Perimeter guarding panels (2 fenced cells)	2 sets
PPE sets — safety shoes, glasses, gloves, hi-vis	70
Welding PPE — helmets, aprons, gauntlets	10
Lockout-tagout (LOTO) kits	12
Fire extinguishers, first-aid stations, eyewash	per facility code

3.8 Group H — Predictive maintenance & condition monitoring · LOW-MEDIUM

Item	Qty
Machinery fault simulator (rotating rig with seeded bearing / misalignment / imbalance faults)	1
Triaxial accelerometer + vibration DAQ set	2
Handheld vibration pen or analyser	3
Current / power monitoring clamps & energy meters	6
Ultrasonic leak detector	2
IIoT gateway + data historian	1 set

3.9 Group I — Software · LOW (*open-source dominates*)

Item	Licence	Seats
ROS 2, Nav2, MoveIt 2, Gazebo, MuJoCo, OpenCV, PCL, PyTorch, LeRobot	Free	30
NVIDIA Isaac Sim / Isaac Lab / Isaac ROS	Free	30
Offline programming (RoboDK, or vendor: RobotStudio / ROBOGUIDE / KUKA.Sim)	Licensed	15
PLC / HMI programming suite (TIA Portal / Studio 5000)	Licensed	12
CMMS / maintenance-management software	Licensed	1 site
Machine-vision suite (Halcon / VisionPro / Keyence) — <i>optional</i>	Licensed	4
MATLAB / Simulink (academic) — <i>optional</i>	Licensed	30
MLOps & experiment tracking (MLflow / W&B)	Free tier	1 team

Item	Licence	Seats
LMS + assessment & attendance platform	Licensed	1

3.10 Group J — Facility · MEDIUM-HIGH

Space / service	Indicative area	Qty
Classroom, 30 seats, power & data at every desk	~60–75 m ²	2
Computer lab, 30 GPU workstations	~75–90 m ²	1
Robotics bench lab (Groups B, D, E)	~100 m ²	1
Controls & maintenance workshop bay (Groups F, H)	~120 m ²	1
Live cell hall — 3-phase power, compressed air, floor loading, hoist access	~250–300 m ²	1
Server & network room with cooling	~15 m ²	1
Store room, tool cribs, lockers	~50 m ²	1
Compressed-air compressor + dryer + reticulation	—	1 set
3-phase power distribution, earthing & panel	—	1 set
HVAC + lighting (≥500 lux at benches)	—	facility
Washroom, pantry, first-aid room	—	facility

3.11 Group K — Technician-tier specific · MEDIUM

Item	Purpose	Ratio	Qty
Fault-injection trainer cell — instructor-switchable faults (sensor disconnect, low air, jam, gripper misalignment, dirty optics, comms drop)	O7, M5 — the core teaching instrument	1 : 15	2
Palletizing / depalletizing rig — robot, pallet stations, case conveyor, layer tooling	O5	—	1
Semi-automatic stretch wrapper + pallets, slip sheets, cases	O5	—	1 + consumables
Sensor alignment & replacement bench kits	M2, O6	1 : 5	6
Hand-tool sets + tool trolleys	M2, M3	1 : 5	6
Torque wrenches, dial indicators, alignment tools	M3	1 : 5	6
Basic metrology kit — gauges, go/no-go, calipers, height gauge	O8	1 : 5	6
Multimeters / clamp meters / insulation testers	M2	1 : 2	15
Tablets — CMMS and SOP retrieval on the floor	M1, M7	1 : 2	15
~~Truck loading/unloading mock-up~~	Deleted in Rev 2	—	—

Why the truck-loading mock-up was deleted: India's dock automation installed base is telescopic conveyors, not robots. Robotic trailer unloading is a 2025–26 global product category with no evidenced Tamil Nadu installations to place graduates against. Dock discipline is taught within O5 on the palletizing rig instead.

3.12 Group L — Consumables & spares (annual) · LOW

Item	Qty
Welding wire, shielding gas, test coupons	consumable lot
Sample workpieces / parts for pick-place & inspection labs	500+ parts
Cable, connectors, DIN rail, terminals, wiring consumables	consumable lot
Consumable spares pool — sensors, cables, vacuum cups, seals, filters, fuses (consumed in training)	1 annual lot
3D printers + filament (fixtures, jigs, gripper fingers)	2 printers
Calibration artefacts (ISO 9283 test rig, checkerboards, reference spheres)	1 set
Spares pool — grippers, cameras, cables, drives	~10% of hardware value

3.13 Capital summary by group

Group	Description	Cost signal	Phase
A	Compute, network & edge	MEDIUM	Launch
B	Learning-tier robotics (bench)	LOW-MEDIUM	Launch
C	Live industrial cells (5, incl. assessment cell)	HIGH	Launch
D	Vision & sensing stock	LOW-MEDIUM	Launch
E	End effectors & tooling	MEDIUM	Launch
F	Industrial controls & integration (<i>2 cohorts</i>)	MEDIUM	Launch
G	Safety infrastructure	MEDIUM	Launch (<i>non-negotiable</i>)
H	Predictive maintenance & condition monitoring	LOW-MEDIUM	Launch
I	Software	LOW	Launch
J	Facility	MEDIUM-HIGH	Launch (<i>or supplied by TNSDC — see the siting note</i>)
K	Technician-tier specific	MEDIUM	Launch
L	Consumables & spares	LOW	Recurring
—	Mobile manipulator + industrial AMR	MEDIUM-HIGH	Year 2

Four observations for the capital plan:

- The graduate tier needs no robots for its first half.** Weeks 1–30 (Diploma) and 1–14 (Training) run on Group A compute and open-source simulation. Group C can be staged.
- The technician tier cannot be staged the same way.** It needs trainer cells from week 3 and live cells from week 11 — but what it needs early is the cheap kit (Groups B and K), not the industrial cells.
- The two tiers are complementary in capex, not competing.** The graduate tier is compute-heavy and hands-light; the technician tier is the reverse and needs **no GPU workstations at all**.
- Group G is not optional and not deferrable**, and Cell 5 is not a luxury — without a ring-fenced assessment cell, the technician tier's headline examination cannot be run as specified.

Sources: [curriculum/syllabus-architecture.md](#) · [curriculum/iti-operator-technician-programme.md](#) · [curriculum/VALIDATION-REPORT.md](#) · [phase-2/lab-procurement-and-partnerships.md](#) · [phase-1/_grounding-reference.md](#) §D · [presentation/deck-brief.md](#) §5–§7. Hour allocations are planning estimates. Named vendors are indicative product classes, not procurement recommendations — confirm current models, availability and pricing at source before any tender. Standards editions (notably ISO 10218) must be verified before printing. Quantities assume zero existing assets; net off anything BGR Neo already holds.